

SUHYEON YU

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EDUCATION

Rice University Master of Computer Science (GPA: 3.91/4.00), Korean Graduate Student Association	Houston, TX, USA Aug. 2025 – Dec. 2026
Sungkyunkwan University B.S. in Computer Science and Engineering	Suwon, South Korea Mar. 2018 – Aug. 2024

WORK EXPERIENCE

Front-End Developer Intern <i>ABC Fitness</i>	Jun. 2026 – Aug. 2026 Austin, TX, USA
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- **Frontend Modernization:** Rewrote the user profile page from scratch in **React**, replacing jQuery as part of a company-wide modernization initiative for ABC Trainerize, which serves **400,000+ fitness professionals** and **1.6M+ clients** worldwide.
- **Async API Coordination:** Implemented asynchronous workflows with **RxJS** that coordinated multi-endpoint profile API requests, suppressed stale responses, and prevented overlapping operations across profile editing and uploading.
- **Unit Testing:** Wrote **Jest** unit tests to verify permission- and role-based access control, conditional rendering, and prevention of overlapping requests during photo/video uploads and profile updates, achieving **97% branch coverage** of profile business logic.
- **UI/UX Redesign:** Redesigned the user profile interface **beyond the original migration scope**, improving information hierarchy through a streamlined profile header, grouped content cards, and modal-based profile editing.
- **Cross-Functional Collaboration:** Collaborated with a UX designer to refine UI details and a QA analyst to verify functional correctness, delivering UX improvements such as drag-and-drop certification reordering to help trainers highlight their top credentials.
- **AI-Assisted Development:** Accelerated development by orchestrating multiple AI agents to build profile sections in parallel in separate Git worktrees and streamlined the development workflow using Figma **MCP** and team-shared agent skills.

Frontend Engineering Intern <i>Sungkyunkwan University</i>	Jan. 2023 – Jan. 2024 Suwon, South Korea
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- **Frontend Development:** Built **Next.js** frontend features for [Online MyeongNyunDang](#), an education platform serving **5,000+ students**, including an admin console integrating REST APIs for course management, homepage curation, and announcement publishing.
- **Initial Load Time Optimization:** Cut homepage initial load time by **~40% (5 to 3 sec)** using server-side rendering and image lazy loading; implemented **on-demand static generation** for announcement detail pages.

PROJECTS

[Orbit Explorer](#) | *React, TypeScript, Three.js, Rust, WebAssembly, D3*

- **3D Visualization:** Built a browser-based orbital digital twin that renders **19,887** tracked orbital objects at **60 fps** using **Three.js** instancing (WebGPU with a WebGL 2 fallback), with a **React** HUD and **D3** panels for laser-link routing, collision screening, and coverage.
- **Parallel Computation:** Implemented an SGP4 orbit propagator in **Rust/WebAssembly** across **4 Web Workers** that write keyframes to a SharedArrayBuffer without per-frame messaging, interpolating every position in **0.25 ms** per frame.
- **Performance Optimization:** Accelerated all-pairs collision screening (198M pairs) by **85×** (280 to 3.3 ms) with a spatial hash, and rebuilt an 18,422-link Starlink laser mesh with Dijkstra routing in **3 ms** to compare latency against fiber.
- **Validation & CI:** Verified the engine against a published reference set (**666/666** states within 0.12 mm) and cross-checked it with satellite.js, running **37** Rust and Vitest tests and daily data snapshots in GitHub Actions.

[Machine Unlearning Comparator](#) | *React, TypeScript, D3, Zustand*

- **AI Interpretability:** Built the frontend end-to-end for an interactive visualization dashboard for evaluating how machine unlearning affects the accuracy, efficiency, and privacy of ML models, earning **100+ GitHub stars**.
- **Data-intensive Visualization:** Implemented **D3/Recharts** visualizations, including **2,000-point** UMAP scatterplots of penultimate-layer activations and **400-point** stacked dot plots that simulate membership inference attacks via a draggable threshold.
- **Interaction Engineering:** Optimized hover, zoom, and drag interactions by replacing full SVG redraws with **D3** joins and in-place updates, cutting per-hover **DOM** manipulations by **99%**.
- **Backend Extension:** Extended the **FastAPI/PyTorch** backend to a face-recognition dataset, enabling another research lab to adopt it.

PUBLICATION

J. Lee, S. Yu, Y. Jang, S. S. Woo, J. Jo. *Unlearning Comparator: A Visual Analytics System for Comparative Evaluation of Machine Unlearning Methods*. **IEEE Transactions on Visualization and Computer Graphics**, vol. 32, no. 3, Mar. 2026.

SKILLS & TOOLS

Frontend: HTML, CSS, React, Next.js, Redux, Zustand, TanStack Query, Tailwind CSS, SCSS, D3, Three.js, WebGPU, WebAssembly
Languages: TypeScript, JavaScript, Python, Rust | **Databases:** SQL (PostgreSQL, MySQL), NoSQL (MongoDB) | **Testing:** Vitest, Jest, pytest
Backend: Node.js, FastAPI, GraphQL, REST APIs | **Cloud & DevOps:** AWS (EC2, S3, Lambda), CI/CD (GitHub Actions), Docker, Git, GitHub